Example 11.21

Evaluate : $\int \frac{(x-1)^2}{x^3+x}\,dx.$

Solution

$$\begin{aligned}\int \frac{(x-1)^2}{x^3+x}\,dx &= \int \frac{x^2+1-2x}{x(x^2+1)}\,dx \\ &= \int\left(\frac{(x^2+1)}{x(x^2+1)}-\frac{2x}{x(x^2+1)}\right)dx \\ &= \int \frac{1}{x}\,dx-2\int \frac{1}{1+x^2}\,dx \\ &= \log|x|-2\tan^{-1}x+c.\end{aligned}$$

Example 11.22

Evaluate : $\int(\tan x+\cot x)^2\,dx.$

Solution

$$\begin{aligned}\int(\tan x+\cot x)^2\,dx &= \int[\tan^2 x+2\tan x\cot x+\cot^2 x]\,dx \\ &= \int[(\sec^2 x-1)+2+(\operatorname{cosec}^2 x-1)]\,dx \\ &= \int(\sec^2 x+\operatorname{cosec}^2 x)\,dx \\ &= \tan x+(-\cot x)+c \\ &= \tan x-\cot x+c.\end{aligned}$$

Example 11.23

Evaluate : $\int \frac{1-\cos x}{1+\cos x}\,dx.$

Solution

$$\begin{aligned}\int \frac{1-\cos x}{1+\cos x}\,dx &= \int \frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}}\,dx=\int \tan^2\frac{x}{2}\,dx \\ &= \int\left(\sec^2\frac{x}{2}-1\right)dx=\frac{\tan\frac{x}{2}}{\frac{1}{2}}-x+c \\ &= 2\tan\frac{x}{2}-x+c.\end{aligned}$$

Example 11.24

Evaluate : $\int\sqrt{1+\sin 2x}\;dx.$

Solution

$$\int\sqrt{1+\sin 2x}\;dx = \int\sqrt{(\cos^2 x+\sin^2 x)+(2\sin x\cos x)}\;dx$$

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